ERRATUM TO "CONDENSATION IN SUBCRITICAL CAUCHY BIENAYMÉ TREES"

IGOR KORTCHEMSKI AND LEONARD VETTER

Abstract. A formal verification in the Lean proof assistant of the results of [I. Kortchemski and L. Vetter, Condensation in subcritical Cauchy Bienaymé trees, Electron. Commun. Probab. 30 (2025), paper no. 79, 1–13, doi:10.1214/25-ECP726] found the errors listed below (the layout and numbering corresponds to the published version).

The statements of the main results, Theorems 1.1 and 3.1, are unaffected; items 4 and 6 repair steps of proofs. The display (2.2), Corollary 4.1 and Proposition 4.5(ii) hold as corrected below, and the claims of Section 2.2 on equivalent laws need the hypothesis $\mu_0 > 0$. The corrected versions of (2.2), Corollary 4.1, Proposition 4.5(ii) and of the classification of Section 2.2 (with $\nu_0 > 0$) are proved in Lean, as are Theorem 3.1, by the argument corrected in item 4, Proposition 4.3, by a variant of the argument of item 6, and the counterexamples of items 1 and 2 except the law ν of item 2. The printed versions of (2.2), Corollary 4.1 and Proposition 4.5(ii) are not refuted in Lean; the reasons given below show that they are false. The corrected criterion of item 1 is checked by hand. The formalization is available at `https://github.com/igor-kortchemski/condensation-subcritical-cauchy-bienayme-trees`, revision `e112dcf`.

Throughout, γ_E denotes Euler's constant, and $\mu_0 = \mu(\{0\})$.

(1) *Section 2.2, p. 4, paragraph after Proposition 2.1.* The criterion "μ is equivalent to a critical offspring distribution if and only if $\lim_{\lambda\to\rho} \lambda F'_\mu(\lambda)/F_\mu(\lambda) \geq 1$" should read as follows. Set $g(\lambda) = \lambda F'_\mu(\lambda)/F_\mu(\lambda)$, which is nondecreasing on $(0, \rho)$. If $\mu_0 > 0$, then μ is equivalent to a critical offspring distribution if and only if either $\lim_{\lambda\uparrow\rho} g(\lambda) > 1$, or $\lim_{\lambda\uparrow\rho} g(\lambda) = 1$ with $\rho < \infty$ and $F_\mu(\rho) < \infty$ (the law μ_ρ defined by $\mu_\rho(k) = \rho^k \mu_k / F_\mu(\rho)$ is then critical). If $\mu_0 = 0$, then μ is equivalent to a critical offspring distribution only when $\mu = \delta_1$.

Reason. For $\mu_0 = \mu_1 = 1/2$ we have $\rho = \infty$ and $g(\lambda) = \lambda/(1 + \lambda) \to 1$, but every law equivalent to μ is supported by $\{0, 1\}$ with positive mass at 0, hence is subcritical.

(2) *Section 2.2, p. 4, last sentence of the same paragraph; the sentence before the list (a)–(e); p. 5, first sentence.* Add the hypothesis $\mu_0 > 0$ to "a supercritical offspring distribution ... is always equivalent to a critical offspring distribution", and the hypothesis $\nu_0 > 0$ to the

classification (a)–(e) and to "when ν is supercritical ... we are in case (a)".

Reason. If $\mu_0 = 0$, every law equivalent to μ has no mass at 0, so its mean is at least 1, with equality only for δ_1. Thus δ_2 is supercritical and not equivalent to a critical law. The law $\nu_0 = 0$, $\nu_n = 6/(\pi^2 n^2)$ for $n \geq 1$, is heavy-tailed with $\alpha = 1$, and every law equivalent to it has mean greater than 1, so none of (a)–(e) applies. Subcritical laws have $\mu_0 > 0$, so the main results are unaffected.

(3) *Section 2.3, p. 5, display (2.2).* The limit is $\mathcal{C}_1 + 1 - \gamma_E$, not $\mathcal{C}_1$:

$$\frac{X_1 + \cdots + X_n - b_n}{a_n} \xrightarrow[n\to\infty]{(d)} \mathcal{C}_1 + 1 - \gamma_E,$$

where $\mathbb{E}[e^{-\lambda \mathcal{C}_1}] = e^{\lambda \log \lambda}$ as in the paper. Equivalently, the limit is $\mathcal{C}_1$ when b_n is replaced by $b_n + (1 - \gamma_E)a_n$.

Reason. The limit is infinitely divisible, with Lévy measure $x^{-2}\,dx$ on $(0, \infty)$, no Gaussian part, and no drift for the truncation $\mathbf{1}_{x \leq 1}$, which matches the centring b_n. Its Laplace exponent is

$$\int_0^\infty \big(e^{-\lambda x} - 1 + \lambda x \mathbf{1}_{x \leq 1}\big) \frac{dx}{x^2} = \lambda \log \lambda + (\gamma_E - 1)\lambda,$$

since the integral equals $\gamma_E - 1$ at $\lambda = 1$. Elsewhere the paper uses (2.2) only through (2.3) and the tightness of $(X_1 + \cdots + X_n - b_n)/a_n$, which a constant shift does not change; only Corollary 4.1 changes (item 5).

(4) *Proof of Theorem 3.1, p. 7, line 1.* Replace $x = \gamma n - 1$ by $x = \gamma n - 2$.

Reason. With $\Delta = (0, 1]$, the event $\overline{W}_n \in (\gamma n - 1, \gamma n]$ is $\{W_n = 0\}$, since W_n is an integer; the conditioning event $\{W_n = -1\}$ is $\overline{W}_n \in (\gamma n - 2, \gamma n - 1]$. As $\varepsilon < \gamma$, we still have $x \geq \varepsilon n$ for n large, and the rest of the proof is unchanged.

(5) *Corollary 4.1, p. 8.* The limit is $-\mathcal{C}_1 - (1 - \gamma_E)$:

$$\frac{\Delta(\mathcal{T}_n) + b_n}{a_n} \xrightarrow[n\to\infty]{(d)} -\mathcal{C}_1 - (1 - \gamma_E).$$

Reason. The paper's derivation from (2.2) and (4.1) is correct given (2.2), and gives this limit once (2.2) is corrected (item 3); replacing n by $n - 1$ in (2.2) changes nothing.

(6) *Proof of Proposition 4.3, p. 9.* The reduction "By Theorem 1.1 (ii), it is enough to show (i) and (ii) with $\mathsf{H}(\mathcal{T}_n)$ replaced by $\mathsf{H}_\star(\mathcal{T}_n)$" holds for (i), since $\mathsf{H}(\mathcal{T}_n) \geq \mathsf{H}_\star(\mathcal{T}_n)$, but not for (ii). For (ii), let E_n be the set of vertices of $\mathcal{T}_n$ that are not descendants of a child of $u_\star(\mathcal{T}_n)$. Then

$$\mathsf{H}(\mathcal{T}_n) \leq |E_n| + 1 + \mathsf{H}_\star(\mathcal{T}_n),$$

since E_n contains $u_\star(\mathcal{T}_n)$ and the ancestors of each of its vertices. Moreover $(|E_n|)_{n \geq 1}$ is tight. Indeed, $|E_n| - 1$ is the last zero of the walk whose steps are the increments of the Łukasiewicz path of $\mathcal{T}_n$ other than the maximal one, read cyclically from it in reverse order.

This walk depends only on the cyclic order of the increments, so it can be computed from $X_1, \dots, X_n$ under $\mathbb{P}(\cdot \mid W_n = -1)$. The maximal increment is unique with probability tending to 1, by Theorem 1.1 (i). Conditionally on this event, the sequence $(X_i^{(n)})$ of Theorem 3.1 is exchangeable and V_n is uniform and independent of it, so these steps have the law of $(X_i^{(n)})$. By Theorem 3.1 they are therefore close in total variation to i.i.d. copies of X, and a random walk with i.i.d. steps of mean $m-1<0$ has finitely many nonnegative values. Given $\eta>0$, choose K with $\mathbb{P}(|E_n|>K)\le\eta$ for n large. Then $\mathbb{P}(\mathsf{H}(\mathcal{T}_n)\ge h_n)\le \eta+\mathbb{P}(\mathsf{H}_\star(\mathcal{T}_n)\ge h_n-K-1)$, and $nQ_{h_n-K-1}\to 0$ because $Q_{j+1}/Q_j\to m>0$ by Lemma 4.4 (i), so the paper's argument for $\mathsf{H}_\star(\mathcal{T}_n)$ applies with h_n-K-1 in place of h_n.

In the same proof, $[\mathcal{T}_n]_{\lceil\Delta_n/2\rceil,\Delta_n}$ has the distribution of $[\mathcal{T}_n]_{1,\lfloor\Delta_n/2\rfloor+1}$ (reverse the order of the children of $u_\star(\mathcal{T}_n)$), which has one tree more than $[\mathcal{T}_n]_{1,\Delta_n-\lfloor\Delta_n/2\rfloor}$ when Δ_n is even. The argument is unchanged, since $\mathbb{P}(\lfloor\Delta_n/2\rfloor+1\le\lfloor\delta n\rfloor)\to 1$ for $\delta=2(1-m)/3$.

Reason. $\mathsf{H}(\mathcal{T}_n)$ is the maximum of the heights of the vertices of E_n and of the vertices of the forest $[\mathcal{T}_n]_{1,\Delta_n}$. Theorem 1.1 (ii) bounds the height of $u_\star(\mathcal{T}_n)$, hence the second part, but not the heights of the other vertices of E_n.

(7) *Proposition 4.5 (ii), p. 11.* Replace $\log(1/m)^{1+\beta}$ by $\log(1/m)^2$: for $\beta\in(0,1)$ and every $\varepsilon>0$,

$$\mathbb{P}\big(|\mathsf{H}(\mathcal{T}_n)-h_n|\ge\varepsilon\log(n)^{1-\beta}\big)\xrightarrow[n\to\infty]{}0,$$

where

$$h_n=\frac{\log(n)}{\log(1/m)}-\frac{c}{(1-\beta)\beta m\log(1/m)^2}\log(n)^{1-\beta}.$$

Reason. The last sentence of the proof (p. 12) correctly obtains $\log(u_n)\sim -c\,n^{1-\beta}/((1-\beta)\beta m\log(1/m)^\beta)$. At $h\sim\log(n)/\log(1/m)$ this gives $\log(u_h)\sim -c\log(n)^{1-\beta}/((1-\beta)\beta m\log(1/m))$, and solving $\log(nQ_h)=0$ as in the case $\beta=1$, as the closing clause of that sentence proposes, divides by one more factor $\log(1/m)$ and gives $\log(1/m)^2$. Part (i) is correct: the two coefficients agree for $\beta=1$. Numerically, iterating (4.5) for $m=0.5$, $c=0.3$, $\beta=0.5$ gives the coefficient 5.00, against 4.995 for the corrected formula and 4.159 for the printed one.

(8) *Misprints.*

- p. 10, second line of the first display: the sum $\sum_{k\ge1}\sum_{j=0}^{k-1}\mu_j s^k$ should read $\sum_{k\ge1}\mu_k\sum_{j=0}^{k-1}s^j$ (the final expression is correct).
- p. 10, after the first display: $\ell(0):=\lim_{s\downarrow0}=0$ should read $\ell(0):=\lim_{s\downarrow0}\ell(s)=0$.
- p. 11, line 1: $c_n\le 1/p$ should read $u_n\le 1/p$.

- p. 11, proof of Proposition 4.5: "for every $\eta \in (0,1)$" should read "for every $\eta \in (0,m)$", as in Lemma 4.4 (ii).